

Introduction to Mathematics and Optimization

May 2025

The references in the suggested answers to this exam set refer to the PDF version *Introduction to Mathematics and Optimization* (Compressed static version (5.12.2024) of interactive book).

The exam answers must be based on the syllabus. Concepts and results in the answer that are not part of the syllabus must be explained using the syllabus.

PROBLEM 1

Let C denote the subset of $\mathbb{R}^2$ given by

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 3, y \leq 3, x + y \geq 4\}$$

and let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = 3x^2 + 2y^2.$$

Consider the optimization problem

$$\min_{(x, y) \in C} f(x, y) \tag{1}$$

(a) State the interior C° and the boundary ∂C of C .

ANSWER:

The interior and boundary of C are, respectively,

$$C^\circ = \{(x, y) \in \mathbb{R}^2 \mid x < 3, y < 3, x + y > 4\} \quad \text{and} \quad \partial C = \{(x, y) \in C \mid x = 3 \vee y = 3 \vee x + y = 4\}.$$

(b) Show that C is compact and that (1) has an optimal solution.

ANSWER:

The optimization problem (1) has an optimal solution by Theorem 5.66, since f is continuous and C is a closed and bounded subset of $\mathbb{R}^2$. Moreover, C is closed and bounded (compact): C is closed because (see Proposition 5.39) it is the intersection of the closed sets

$$\begin{aligned} &\{(x, y) \mid x \leq 3\} \\ &\{(x, y) \mid y \leq 3\} \\ &\{(x, y) \mid x + y \geq 4\}. \end{aligned}$$

Each of these subsets is closed because they have the form $\{(x, y) \in \mathbb{R}^2 \mid g(x, y) \leq 0\} = g^{-1}((-\infty, 0])$, where g is a continuous function (see Proposition 5.42 and Proposition 5.51). That C is bounded follows from Definition 5.29, since

$$(x, y) \in C \implies x^2 + y^2 \leq 3^2 + 3^2 = 18$$

(FM elimination implies $x \geq 1$ and $y \geq 1$). Thus $C \subseteq B(0, 5)$.

- (c) Show that if $(x, y) \in C$ is an optimal solution to (1), then $(x, y) \in \partial C$.

ANSWER:

If $(x, y) \in C$ is an optimal solution to (1) and $(x, y) \in C^\circ$, then (x, y) would be a critical point for f by Proposition 7.21, but this is impossible, since $\nabla f(x, y) = (6x, 4y)$ is not equal to $(0, 0)$ for $(x, y) \in C^\circ$. Therefore $(x, y) \in \partial C$.

- (d) Show that (1) is a convex optimization problem with a unique solution.

ANSWER:

We need to show that f is a convex function and C a convex subset of $\mathbb{R}^2$. That C is a convex subset follows from the fact that it is the intersection (see Exercise 4.23) of the three convex subsets $x \leq 2$, $y \leq 2$ and $x + y \geq 3$. These subsets are convex because each has the form $\{(x, y) \in \mathbb{R}^2 \mid g(x, y) \leq 0\}$, where g is a convex function (see Lemma 4.27). That f is a convex function follows from Theorem 8.23, since

$$\nabla^2 f(x) = \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix}$$

is positive definite by Exercise 8.28. In fact, this implies that f is a strictly convex function again by Theorem 8.23. Therefore the optimization problem is convex and has a unique solution by Theorem 6.13.

- (e) Write down the Karush-Kuhn-Tucker conditions for (1).

ANSWER:

The KKT conditions are, according to (9.24),

$$\begin{aligned} \lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ \lambda_3 &\geq 0 \\ x - 3 &\leq 0 \\ y - 3 &\leq 0 \\ -x - y + 4 &\leq 0 \\ \lambda_1(x - 3) &= 0 \\ \lambda_2(y - 3) &= 0 \\ \lambda_3(-x - y + 4) &= 0 \\ 6x + \lambda_1 - \lambda_3 &= 0 \\ 4y + \lambda_2 - \lambda_3 &= 0 \end{aligned}$$

- (f) Solve the optimization problem (1) using a method from the syllabus that is based on $C = C^\circ \cup \partial C$.

ANSWER:

We solve the optimization problem based on Proposition 7.44. We already know from the previous part that the optimal solution lies on the boundary

$$\partial C = \{(x, y) \in C \mid x = 3 \vee y = 3 \vee x + y = 4\}$$

of C :

Consider the three cases one at a time:

1) $x = 3$:

This reduces to minimizing $18 + 2y^2$ with the constraint $1 \leq y \leq 3$. The critical point of $18 + 2y^2$ is $y = 0$, which does not satisfy $1 \leq y \leq 3$. Therefore the minimum here is $y = 1$, i.e. the minimum is $(3, 1)$ for this part of the boundary. Here $f(3, 1) = 29$.

2) $y = 3$: This reduces to minimizing $3x^2 + 18$ with the constraint $1 \leq x \leq 3$. The critical point of $3x^2 + 18$ is $x = 0$, which does not satisfy $1 \leq x \leq 3$. Therefore the minimum here is $x = 1$, i.e. the minimum is $(1, 3)$ for this part of the boundary. Here $f(1, 3) = 21$.

3) $x + y = 4$:

This reduces to minimizing $3x^2 + 2(4 - x)^2 = 5x^2 - 16x + 32$ with the constraint $1 \leq x \leq 3$. The critical point of $3x^2 - 12x + 18$ is the solution to $10x - 16 = 0$. Therefore the minimum here is $x = \frac{8}{5}$, i.e. the minimum is $(\frac{8}{5}, \frac{12}{5})$ for this part of the boundary. Here $f(\frac{8}{5}, \frac{12}{5}) = \frac{96}{5}$.

Thus the optimal solution is $(\frac{8}{5}, \frac{12}{5})$, since $\frac{96}{5} < 21$ and $\frac{96}{5} < 29$.

PROBLEM 2

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + xy + y^2 + xz + z^2 + yz$$

for $(x, y, z) \in \mathbb{R}^3$.

(a) Find a symmetric 3×3 matrix A such that

$$f(x, y, z) = \begin{pmatrix} x & y & z \end{pmatrix} A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

for all $(x, y, z) \in \mathbb{R}^3$.

ANSWER:

Via matrix multiplication we obtain

$$\begin{aligned} \begin{pmatrix} x & y & z \end{pmatrix} A \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & d & e \\ d & b & g \\ e & g & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} ax + dy + ez \\ dx + by + gz \\ ex + gy + cz \end{pmatrix} \\ &= x(ax + dy + ez) + y(dx + by + gz) + z(ex + gy + cz) \\ &= ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2gyz. \end{aligned}$$

Thus we can use $a = b = c = 1$ and $d = e = g = \frac{1}{2}$. That is,

$$A = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}.$$

(b) Show that A is positive semidefinite.

ANSWER:

First we reduce A to a diagonal matrix by using Theorem 8.29 and the algorithm in Section 8.6.

We first multiply column 1 by $-\frac{1}{2}$ and add it to columns 2 and 3. Then we multiply row 1 by $-\frac{1}{2}$ and add it to rows 2 and 3. This transforms the matrix into

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{4} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{3}{4} \end{pmatrix}.$$

We now multiply column 2 by $-\frac{1}{3}$ and add it to column 3. Then we multiply row 2 by $-\frac{1}{3}$ and add it to row 3. This gives us the diagonal matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{2}{3} \end{pmatrix}.$$

This diagonal matrix is positive semidefinite according to Exercise 8.28. It now follows from Proposition 3.41 that A is positive semidefinite. In fact it is positive definite, since the diagonal entries are positive.

(c) Show that $f(x, y, z) \geq 0$ for all $(x, y, z) \in \mathbb{R}^3$.

ANSWER:

From the previous parts we have that

$$f(x, y, z) = \begin{pmatrix} x & y & z \end{pmatrix} A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

and that A is positive semidefinite. According to the definition of positive semidefinite matrices in 3.7.1, $f(x, y, z) \geq 0$ for all $(x, y, z) \in \mathbb{R}^3$.

(d) Find all $(x, y, z) \in \mathbb{R}^3$ such that $f(x, y, z) = 0$.

ANSWER:

In fact we showed in a previous part that A is positive definite. Therefore $f(x, y, z) = 0$ if and only if $(x, y, z) = (0, 0, 0)$ according to the definition of positive definite matrices in 3.7.1.

(e) Is f a strictly convex function?

ANSWER:

The Hessian matrix of f is $2A$, which is positive definite since A is positive definite. Therefore f is a strictly convex function according to Theorem 8.23.

(f) Show that $(0, 0, \frac{1}{2})$ is a solution to the optimization problem

$$\begin{aligned} \min f(x, y, z) \\ x + y + 2z = 1 \end{aligned}$$

by using the KKT conditions as they are stated in the syllabus. Do the KKT conditions have a unique solution?

ANSWER:

We rewrite the optimization problem as

$$\begin{aligned} \min f(x, y, z) \\ x + y + 2z - 1 \leq 0 \\ -x - y - 2z + 1 \leq 0. \end{aligned}$$

This optimization problem is convex and has the KKT conditions

$$\begin{aligned} \lambda_1, \lambda_2 &\geq 0 \\ \lambda_1(x + y + 2z - 1) &= 0 \\ \lambda_2(-x - y - 2z + 1) &= 0 \\ 2x + y + z + \lambda_1 - \lambda_2 &= 0 \\ x + 2y + z + \lambda_1 - \lambda_2 &= 0 \\ x + y + 2z + 2(\lambda_1 - \lambda_2) &= 0 \end{aligned}$$

For $(x, y, z) = (0, 0, \frac{1}{2})$ the KKT conditions are satisfied with $\frac{1}{2} = \lambda_2 - \lambda_1$ and $\lambda_1, \lambda_2 \geq 0$. Conclusion via Theorem 9.34(ii). Here one sees that the KKT conditions do not have a unique solution.

PROBLEM 3

Let $C \subseteq \mathbb{R}^3$ be the set of points $(x, y, z) \in \mathbb{R}^3$ that satisfy the following inequalities:

$$\begin{aligned}x + y - z &\leq 1 \\ -x + 2y + z &\leq 2 \\ x - y + z &\leq 3 \\ -x - y - z &\leq 4.\end{aligned}$$

(a) Use Fourier-Motzkin elimination to eliminate z from the inequalities.

ANSWER:

We start from the inequalities

$$\begin{aligned}x + y - 1 &\leq z \\ z &\leq 2 + x - 2y \\ z &\leq -x + y + 3, \\ -x - y - 4 &\leq z\end{aligned}$$

and obtain by Fourier-Motzkin elimination the following inequalities in x and y :

$$\begin{aligned}x + y - 1 &\leq 2 + x - 2y, \\ x + y - 1 &\leq -x + y + 3, \\ -x - y - 4 &\leq -x + y + 3, \\ -x - y - 4 &\leq 2 + x - 2y,\end{aligned}$$

or collectively

$$\begin{aligned}y &\leq 1 \\ x &\leq 2, \\ -\frac{3}{2} &\leq y, \\ -6 &\leq 2x - y,\end{aligned}$$

(b) Next eliminate y from the inequalities you obtained in (a).

ANSWER:

The inequalities from (a) can be written as $x \leq 2$ together with

$$\begin{aligned}y &\leq 1 \\ -\frac{3}{2} &\leq y, \\ y &\leq 2x + 6\end{aligned}$$

which yield $-\frac{15}{4} \leq x$. Altogether we have

$$-\frac{15}{4} \leq x \leq 2 \quad \text{and} \quad -\frac{3}{2} \leq y \leq 1$$

(c) Show from (b) and (a) that C is bounded.

ANSWER:

The inequality $x + y - 1 \leq z \leq 2 + x - 2y$ from (a), together with the bounds on x and y from (b), gives

$$-\frac{25}{4} = -\frac{21}{4} - 1 \leq z \leq 2 + 2 + 3 = 7$$

Hence C is bounded according to Definition 5.29, since

$$(x, y, z) \in C \implies \sqrt{x^2 + y^2 + z^2} \leq \sqrt{4^2 + 2^2 + 7^2} < 9$$

i.e. $C \subseteq B(0, 9)$.

(d) Show that C is compact.

ANSWER:

From (b) we now know that C is bounded, so according to Definition 5.43 we still need to show that C is closed. This follows from

$$C = f_1^{-1}(J) \cap f_2^{-1}(J) \cap f_3^{-1}(J) \cap f_4^{-1}(J),$$

where

$$\begin{aligned} f_1(x, y, z) &= x + y - z - 1 \\ f_2(x, y, z) &= -x + 2y + z - 2 \\ f_3(x, y, z) &= x - y + z - 3 \\ f_4(x, y, z) &= -x - y - z - 4 \end{aligned}$$

and $J = (-\infty, 0]$. Here we have (as usual) used Proposition 5.42, Proposition 5.51 and Proposition 5.39 as well as the fact that the functions $f_1, f_2, f_3, f_4 : \mathbb{R}^3 \rightarrow \mathbb{R}$ are continuous.

(e) Let the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by

$$f(x, y, z) = 12 \cos(\sin(e^{x^2-y^3}) + y^{17}) + 15xy - e^{x^2-5x+6}.$$

Does the optimization problem

$$\begin{aligned} \min f(x, y, z) \\ (x, y, z) \in C. \end{aligned}$$

have a solution? Justify your answer.

ANSWER:

The function f is continuous via composition, sums and products of continuous functions. Since C is compact it follows from Theorem 5.66 that the optimization problem has a solution.